Prince Mathew

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Career Profile

I am a postdoctoral researcher in the Formal Methods and Verification group at Université Libre de Bruxelles, working on strategy synthesis for partially observable Markov decision processes. My research lies at the intersection of automata theory, formal verification, learning, and synthesis. During my PhD at IIT Goa and my subsequent research fellowship there, I focused on the learning and equivalence of one-counter systems. More broadly, I develop efficient algorithms with formal correctness guarantees, with an emphasis on active automata learning, equivalence checking, synthesis, and implementable verification methods.

Publications

- **Synthesizing POMDP Policies: Sampling Meets Model-checking via Learning** [arXiv Link] — *Jul 26, 2026*
  Co-authors: Dr. Debraj Chakraborty, Dr. Anirban Majumdar, Dr. Sayan Mukherjee, Prof. Jean-François Raskin. — *CAV 2026 (to appear)*
- **Edit Distance of Finite-Valued Transducers** [Link] — *Jul 7, 2026*
  Co-authors: Dr. Saina Sunny. — *ICALP 2026*
- **Scalable Learning of One-Counter Automata via State-Merging Algorithms** [Link] — *Sep 8, 2025*
  Co-authors: Dr. Shibashis Guha, Dr. Anirban Majumdar, Dr. Sreejith A.V. — *FSTTCS 2025*
- **Learning Deterministic One-Counter Automata in Polynomial Time** [Link] — *Mar 7, 2025*
  Co-authors: Dr. Vincent Penelle, Dr. Sreejith A.V. — *LICS 2025*
- **Learning Real-time One-Counter Automata Using Polynomially Many Queries** [Link] — *May 1, 2025*
  Co-authors: Dr. Vincent Penelle, Dr. Sreejith A.V. — *TACAS 2025*
- **Equivalence of Deterministic Weighted Real-time One-Counter Automata** [Link] — *Feb 3, 2025*
  Co-authors: Dr. Vincent Penelle, Dr. Prakash Saivasan, Dr. Sreejith A.V. — *ICLA 2025*
- **Weighted One-Deterministic-Counter Automata** [Link] — *December 12, 2023*
  Co-authors: Dr. Vincent Penelle, Dr. Prakash Saivasan, Dr. Sreejith A.V. — *FSTTCS 2023*
- **CAP: A Cellular Automata Based Fuzzy Classifier** [Link] — *February 26, 2022*
  Co-authors: Dr. Abdul Nizar M — *Materials Today Proceedings*
- **Optical Music Recognition Using Image Processing and Machine Learning** [Link] — *August 11, 2018*
  Co-authors: Rahul Vijayakumar, Aju Tom Kuriakose, Jesmy Sunny, Dr. Ramani Bai V — *IJCSE 2018*

Experience

- **Université Libre de Bruxelles** — *November 2025 to present*
  Postdoctoral Researcher
- **Indian Institute of Technology Goa** — *January 2025 to October 2025*; *January 2024 to July 2024*; *February 2018 to December 2018*
  Junior Research Fellow
- **Tata Elxsi, Trivandrum** — *January 2017 to January 2018*
  Senior Engineer

- **Central Polytechnic College, Trivandrum** *June 2016 to October 2016*
  Guest Lecturer

Funded Projects

- **FNRS Chargé de Recherches (CR) Fellowship** *October 2026 to September 2029*
  Learning and Equivalence of Automata with Resources (LEARn)

Invited and Conference Talks

- Synthesizing POMDP Policies: Sampling Meets Model-checking via Learning, AQUARIUM 2026 *March 24, 2026*
- Learning Deterministic One-Counter Automata, RP 2025 *October 2, 2025*
- Learning Deterministic One-Counter Automata in Polynomial Time, LICS 2025 *June 24, 2025*
- Learning Real-Time One-Counter Automata Using Polynomially Many Queries, TACAS 2025 **[Talk]** *May 6, 2025*
- Weighted One-Deterministic-Counter Automata (Lightning Talk), ACM ARCS 2025 *Feb 27, 2025*
- Equivalence of Deterministic Weighted Real-time One-Counter Automata, ICLA 2025 *Feb 3, 2025*
- Learning Real-Time One-Counter Automata Using Polynomially Many Queries, RHPL 2024 **[Talk]** *Dec 18, 2024*
- Learning one-counter automata using SAT solver, Highlights 2024 **[Talk]** *Sep 19, 2024*
- Weighted one deterministic-counter automata, FSTTCS 2023 **[Talk]** *July 2023*
- One deterministic-counter automata, Highlights 2023 **[Talk]** *July 2023*
- Weighted one-deterministic-counter automata, FM Update Meeting 2023 **[Talk]** *June 2023*

Other Talks

- Synthesizing POMDP Policies: Sampling Meets Model-checking via Learning, CFV Seminar, ULB, 2026 *March 6, 2026*
- Learning Deterministic One-Counter Automata in Polynomial Time, IIT Bombay *July 18, 2025*
- Learning Real-Time One-Counter Automata Using Polynomially Many Queries, TIFR Mumbai *April 4, 2025*
- Learning One-Counter Automata Using SAT Solver, Université Libre de Bruxelles, Belgium *September 10, 2024*
- Learning One-Counter Automata Using SAT Solver, Université de Mons, Belgium *September 9, 2024*
- Learning One-Counter Automata Using SAT Solver, RWTH Aachen University, Germany *September 6, 2024*

Community Services

- Sub-reviewer: International Symposium on Formal Methods (FM), 2026
- Sub-reviewer for the IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS), 2025
- Sub-reviewer for the IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS), 2024
- Sub-reviewer: International Conference on Concurrency Theory (CONCUR), 2023
- Reviewed NPTEL transcripts for the course “Theory of Computation” by Prof. Somenath Biswas

🏛 Awards and Recognitions

- Invited participant, Autobóz Research Camp 2026 *July 2026*
- Received invitation from ACM India to present our work at ARCS 2026 *December 2025*
- Received grant from Google for our work on learning one-counter automata *May 2025*
- Received ACM/IARCS travel grant for attending LICS 2025 *May 2025*
- Selected for scholarship for the Logic Mentoring Workshop at LICS 2025 *April 2025*
- Awarded with the ETAPS 2025 scholarship for participating in TACAS 2025 *April 2025*
- Awarded with the ACM/IARCS travel grant for attending TACAS 2025 *April 2025*
- Received travel grant from ACM India for attending ARCS 2025 *March 2025*
- Received invitation from ACM India to present our work at ARCS 2025 *December 2024*
- Received grants from RWTH Aachen University, Germany and University of Antwerp, Belgium for research visits *September 2024*

🏛 Research Visits

- Tata Institute of Fundamental Research (TIFR), Mumbai, India *July 16 to July 22, 2025*
- Tata Institute of Fundamental Research (TIFR), Mumbai, India *March 24 to April 4, 2025*
- University of Antwerp, Belgium *Sep 9 to 13, 2024*
- RWTH Aachen University, Germany *Sep 4 to 6, 2024*
- The Institute of Mathematical Sciences, HBNI, India *June 1 to 30, 2022*
- Chennai Mathematical Institute, Chennai, India *January 6 to May 30, 2020*

🎙 Teaching Experience

- Guest Lecturer, *Software Testing and Beyond Summer School, UAntwerpen* *Summer 2026*
  Invited to deliver a two-day session on model learning.
- Embedded Systems Design *Winter 2025 – 26*
- Introduction to Computing *Spring 2020 – 21, Autumn 2023 – 24*
- Randomised Algorithms *Spring 2022 – 23*
- Software Tools *Spring 2022 – 23*
- Data Structures and Algorithms *Autumn 2022 – 23, Summer 2022, Autumn 2021 – 22, Autumn 2020 – 21*
- Automata Theory *Spring 2020 – 21, Spring 2018 – 19*
- Logic in Computer Science *Autumn 2019 – 20*
- Advanced Algorithms *Spring 2018 – 19*

Education

- **Indian Institute of Technology Goa** *January 2019 to December 2024*
  Ph.D. in Theoretical Computer Science CPI: 8.45
  - **Chennai Mathematical Institute, Tamil Nadu** *January 2020 to April 2020*
    Ph.D. Coursework

- **College of Engineering Trivandrum, Kerala** *August 2014 to April 2016*
  M.Tech in Computer Science and Engineering CGPA: 9.11
- **Saintgits Engineering College, Kerala** *July 2010 to April 2014*
  B.Tech in Computer Science and Engineering CGPA: 7.62
- **Don Bosco Higher Secondary School, Kottayam, Kerala** *2008 to 2010*
  Class 12 Aggregate: **86.36** %
- **Don Bosco Higher Secondary School, Kottayam, Kerala** *2007 to 2008*
  Class 10 CGPA: **10/10**

Other Activities

- Volunteered and contributed towards the successful organisation of FSTTCS 2021 and FSTTCS 2022
- Volunteered in organising Formal Methods Update Meeting 2023
- Volunteered in organising the Indian School on Logic and its Applications 2024
- PG representative in the Senate Students Advisory Committee, IIT Goa from 2019 to 2021
- Member of the evaluation team for the national level technical project exhibition - SRISHTI 2025
- Successfully cleared the UGC NET examination and have met the eligibility criteria for lectureship
- Passed BEC Vantage conducted by the University of Cambridge
- Passed grade 5 vocals with merit conducted by Trinity College London
- Passed grade 3 piano with merit conducted by Trinity College London

Projects

- **CPLUS** *November 2025 to April 2026*
  A tool for strategy synthesis for partially observable Markov decision processes.
  - Tools & technologies used: AALpy, Python
  - This tool is associated with our paper "Synthesizing POMDP Policies: Sampling Meets Model-checking via
Jul 26, 2026 Learning" that will appear in CAV 2026. Code: **zenodo.org/records/19850366**
- **OCA-L*** *June 2025 to August 2025*
  A tool for active learning of one-counter automata using RPNI.
  - Tools & technologies used: AALpy, DFAMiner, Python
  - This tool is associated with our paper "Scalable Learning of One-Counter Automata via State-Merging Algorithms " that appeared in FSTTCS 2025. Code: **zenodo.org/records/17310292**
- **MinOCA** *January 2024 to December 2024*
  A tool for active learning of one-counter automata using polynomially many queries
  - Tools & technologies used: DFAMiner, Python
  - This tool is associated with our paper "Learning Real-Time One-Counter Automata Using Polynomially Many Queries" that appeared in TACAS 2025. Code: **zenodo.org/records/14604419**
- **MoES - River Width Extraction Project** *February 2018 to December 2018*
  Web application for MoES for calculating river water discharge from satellite images.

- Tools & technologies used: Python, Django
- This work was done as a part of a joint project with Dr. Sreejith A.V (IIT Goa) and Dr. Gaurav Kumar (IISER Bhopal).

- **Skill Center - Tata Elxsi** *January 2017 - February 2018*

  Web application to manage employee profiles of Tata Elxsi.

  - Tools & technologies used: C#, MVC, .NET
  - The application manages skills, project allocation, training, funnel management, performance evaluations, and resume creation. It helps assign projects based on employee skills, plan and track training, search for employees, assess performance, and create resumes.

- **Fuzzy Classifier using Continuous Cellular Automata [Report][Code]** *June 2015 - September 2016*

  Continuous cellular automata-based fuzzy classifier in Java using Weka.

  - Tools & technologies used: Java, Weka
  - This tool is associated with our paper "CAP: A Cellular Automata Based Fuzzy Classifier".
  - Project done as part of Masters degree thesis.

- **Open Image Transcriptor [Report][Code]** *June 2013 - June 2014*

  Image processing tool to recognise musical notes from sheet music and convert it into a MIDI file.

  - Tools & technologies used: Java, Python
  - This tool is associated with our paper "Optical music recognition using image processing and machine learning".
  - Project done as part of Undergraduate project.